

SUYOG HANMANT BARGULE

AI Engineer (Agentic AI & Computer Vision)

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Profile Summary

AI & Computer Vision Engineer with 4+ years of experience in building real-time, production-grade AI systems across surveillance, traffic monitoring, assistive technology, and generative AI. Specialized in computer vision optimization, agentic AI, RAG pipelines, and scalable model deployment. Proven impact with up to 81% accuracy improvement and 80% reduction in manual effort through automation.

Work Experience

Simplifi Dentistry Solutions India Pvt Ltd , Lucknow (Remote)– AI Engineer

📅 Oct 2025-Jan 2026

- Built an AI automation tool using RAG, LangFlow, Playwright, and LangChain.
- It helps dental clinics collect patient insurance info, verify benefits, and manage claim processing automatically.

Futops Technologies, Pune – AI & Computer Vision Engineer

📅 Oct 2022-Aug 2025

- Developed an agentic AI-based lead generation system using n8n, Apollo AI, Afify Scraper, and LinkedIn/company data to automatically identify, analyze, and qualify leads. Automated personalized outreach and follow-up emails, reducing manual prospecting efforts by 80% and improving conversion efficiency.
- Built a local LLM-powered chatbot using LangChain and DeepSeek (14B) integrated via Ollama with CUDA support, and deployed it on the company website to enable low-latency, private inference.
- Designed and implemented a computer vision-based traffic surveillance system to detect helmet violations, triple riding, and red-light jumping, achieving an 81% improvement in detection accuracy. Introduced a best-frame (snapshot) selection technique, increasing object detection accuracy from 40% to 75%. (SPOT Award)
- Developed a real-time assistive system for visually impaired users using AWS cloud services, achieving 82% accuracy through a scalable, cloud-native AI architecture optimized for real-time inference.
- Optimized a face recognition pipeline to achieve 95% recognition accuracy and 90% face detection accuracy, including age, gender, and mask detection, with inference latency of 30–35 ms, enabling edge AI deployment in low-latency environments.
- Implemented panoramic 360° CCTV imaging and low-light image stitching solutions using OpenCV and GStreamer, improving surveillance coverage and image clarity. (SPOT Award)

AI4SEE Pvt. Ltd, Bangalore – AI Engineer

📅 Feb 2021-Sep 2022

- Built a Flutter-based rope jump counter using a TensorFlow Lite model for real-time analysis.
- Developed a mobile app for direct farmer-to-retail sales; boosted sales by 20% with a regression model predicting prices five days in advance.

Skills

- **Programming Languages:** Python, C++
- **Machine Learning & Deep Learning:** Machine Learning Engineer skills, Deep Learning, Supervised & Unsupervised Learning, CNNs, GANs, Vision Transformers, NLP, Multi-modal AI
- **GenAI, LLMs & Agentic AI:** Large Language Models (LLMs), LLaMA, DeepSeek, Hugging Face, LangChain, RAG, Agentic AI, LangGraph, LangSmith, LangFlow, Prompt Engineering, LoRA Fine-tuning, Stable Diffusion, Synthetic Data Generation
- **Computer Vision:** YOLO, Object Detection, Image Segmentation, Image Processing, Performance Optimization, Face Recognition, Low-Light Imaging, Panoramic Imaging
- **Libraries & Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, ONNX, OpenVINO, CUDA, NumPy, Pandas
- **MLOps, CI/CD & Model Deployment:** MLOps, Model Deployment, Model Monitoring, CI/CD Pipelines, Data Pipelines, ONNX Runtime, TensorRT, FastAPI, REST Full APIs, Docker, Kubernetes,
- **Cloud & Infrastructure:** AWS (Rekognition, S3, EC2, SageMaker), Azure, Linux, Kafka, ZeroMQ, GStreamer, OpenGL, Playwright
- **Databases & Vector Stores:** SQL, PostgreSQL, MongoDB, Qdrant, Pinecone, FAISS
- **Soft Skills:** Teamwork & Collaboration, Adaptability, Analytical Thinking, Problem-Solving, Effective Communication

Languages

English, Hindi, Marathi

Education

Master of Computer Application

Shivaji University Computer Science Department | Graduation Year: 2019

Certifications

Data Science and Machine Learning, ExcelR Solutions, Jan 2019 – May 2019

PyTorch for Deep Learning Bootcamp, Udemy – Andrei Neagoie & Daniel Bourke, Jun 2024 – May 2025